Heinrich Heine Universität Düsseldorf

Report on the Research Internship: Investigating RAFT-Polymerization and PISA on Poly(Lipoates)

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1 Introduction

1.1 Lipoic Acid

1.1.1 General Background

1.1.2 Esterification

1.2 RAFT

1.2.1 General Background

1.2.2 Application to Lipoates

1.3 PISA

1.3.1 General Background

1.3.2 Reverse PISA

1.3.3 Degradation of Poly(Lipoate)

2 Experimental

2.1 Methods

Within this work a variety of methods have been used. In chapter 1, some of the theoretical background for the reactions has been displayed, whereas the focus lies on the analytical aspects in this chapter. Different instruments have been used for monitoring the reaction course, as well as the chemical and physical properties of the monomers and polymers. The following sections give more insight on the selected methods and for some of them a description of the handling of the analysis data to obtain or display results. For writing this report, the translation tool *DeepL Translator* has been used for formulaic purposes.[1]

2.1.1 SEC

Size-exclusion chromatography (SEC) is an analysis method to characterize polymers in regard to their molecular weight. This provides information on the molar mass distribution and characteristic values, such as the number average molar mass (M_n), mass average molar mass (M_w) and the dispersity ($Đ$).
The measurements for the SEC analysis were performed on the instruments within the Chair of Macromolecular Chemistry at Heinrich-Heine University (HHU) according to the established procedure within the group. The majority of the measurements have been collected on the instrument with *N,N*-dimethylformamide (DMF) as a solvent, the remaining measurements have been made on the tetrahydrofuran (THF) instrument. The data has been exported as an ASCII file and plotted on OriginPro 2025 v10.2.

2.1.2 NMR

The technique nuclear magnetic resonance (NMR) spectroscopy can be used to determine the chemical composition, and by that also the purity, of chemicals, to obtain the conversion of a and monitor the polymerization reaction.
For the analysis, ^{1}*H*-NMR measurements were performed by the Center for Molecular and Structural Analytics @ Heinrich-Heine University (CeMSA@HHU) on the 600 MHz spectrometer. The results have been displayed and analyzed via the software MestReNova v17.0.1 which has been obtained through the chemistry department at Heinrich Heine University.

2.1.2.1 DOSY

Diffusion-ordered spectroscopy (DOSY) is a form of NMR spectroscopy wherein the different signals are measured as a function of their chemical shift as well as their diffusion coefficient (D). This allows the differentiation between different molecules within a mixture which makes it a powerful analysis for analyzing block copolymers. Within the CeMSA@HHU, the measurments have been performed on the 600 MHz Evo Neo spectrometer. For the analysis, the software TopSpin 5.0.0 Processing by Bruker has been used. The software can be acquired through the academic processing license on the Bruker website.[2]
The analysis of DOSY spectra can be done the following way:

1. Make a copy of the folder with the raw data
2. Navigate to "*/pdata/1*" within the copied raw data folder and localize the file called "*2rr*"
3. Open TopSpin
4. Drag and drop the file "*2rr*" into TopSpin
5. Enter the following commands into the command line after the previous one is done:
    a) xf2
    b) abs2
    c) eddosy
    d) setdiffparm
    e) dosy2d
6. TopSpin can be closed once the commands have been run and MestReNova can be opened
7. The edited folder with the data can be dragged and dropped into MestReNova
8. In the *Stacked* tab, the button *DOSY Plot* needs to be pressed
9. The display of the spectra can be edited via the *Properties* window as usual

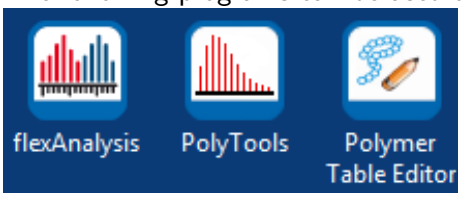
2.1.3 MALDI-ToF-MS

In order to receive information on the end groups of polymers, matrix assisted laser desorption/ionization time of flight mass spectrometry (MALDI-ToF-MS) can be measured by the CeMSA@HHU. For this, the samples need to be in solution and an approximate molar mass is needed to set the measurement window. The concentration should be approximately 1 mg mL^{-1} and for the samples within this research work, THF has been selected as a solvent.
The analysis of the MALDI-ToF-MS spectra can be done as follows:

1. Request the measurement data from the CeMSA@HHU. This can be done via mail with specification of the measurement number which can be found on the handout result sheet

2. Copy the measurement data onto a USB stick and access the mass spectrometry computer within the Chair of Macromolecular Chemistry at HHU

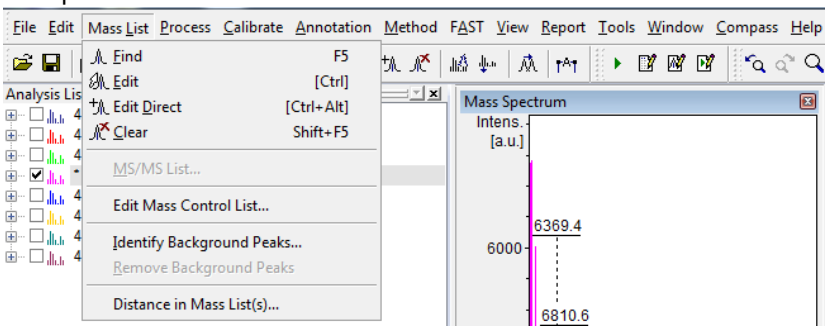
3. The following programs can be used to analyze MALDI-ToF-MS data of polymers:



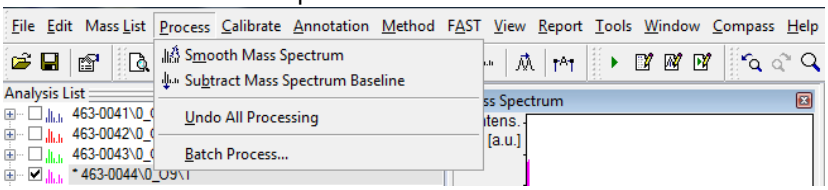
   a) flexAnalysis: Edit the peaklist

      i. Open your measurement data under *File – Open*

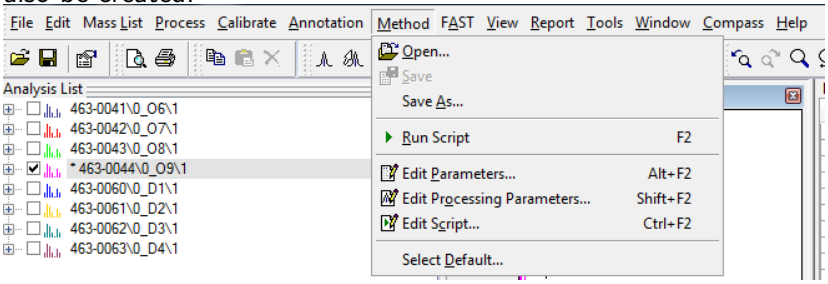
      ii. The peak list can be edited in the tab *Mass Liss*.



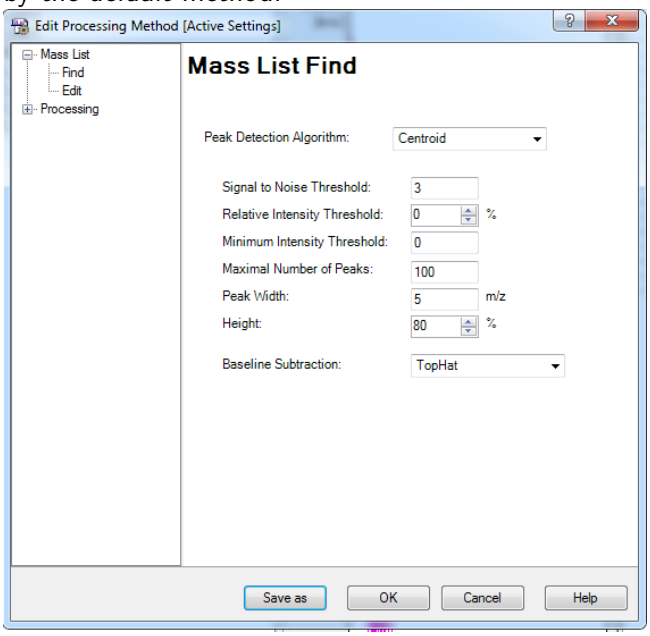
      iii. In the tab *Process* the spectra can be smoothed and a baseline can be substracted.



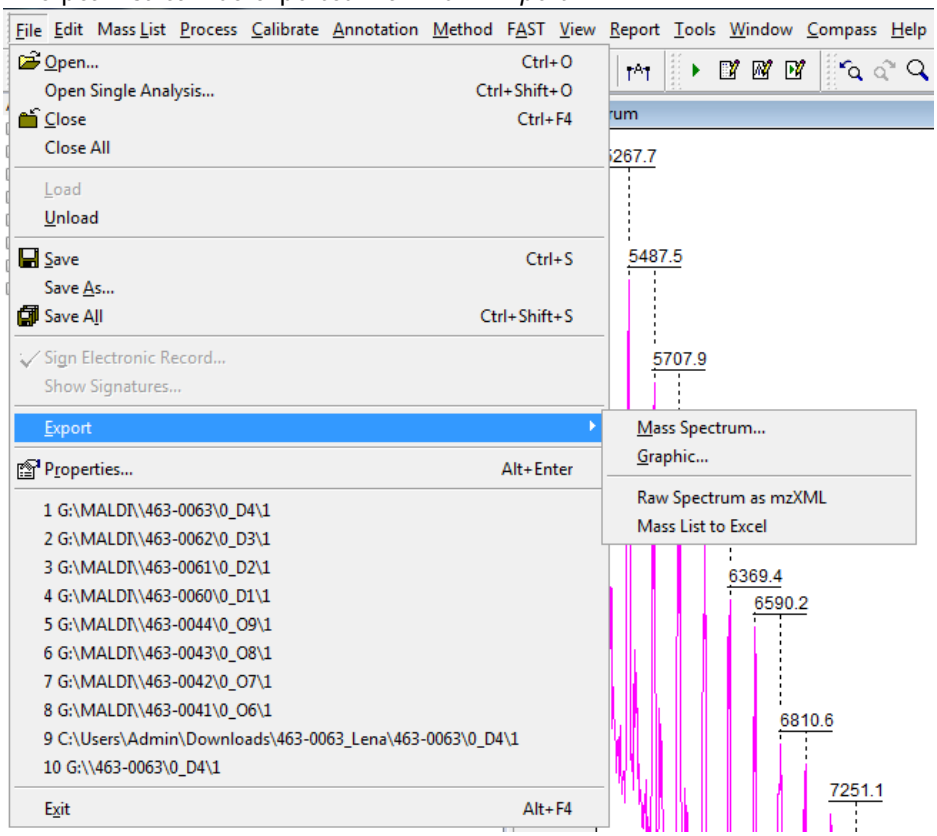
      iv. The parameters of the peak detection can be edited via *Method*. A script for analysis can also be created.



v. Within *Method – Edit Processing Method* different settings for detecting signals can be edited. This can be useful if many signals are in a narrow range and cannot be recognized by the default method.



vi. The peaklist can be exported via *File – Export*

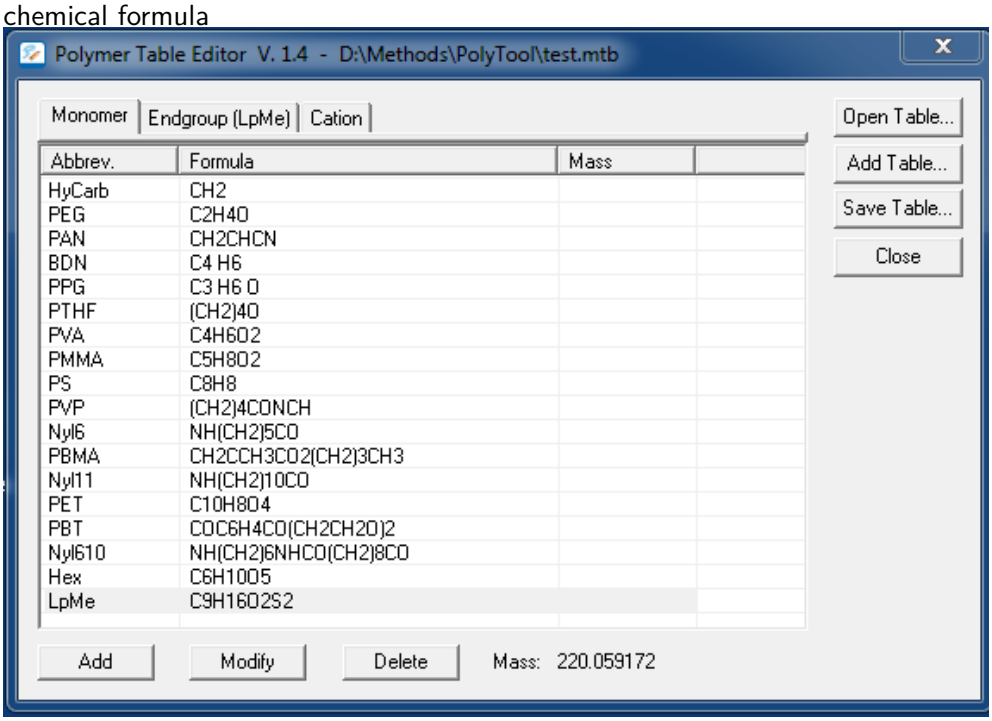


vii. Save your edits under *File – Save*

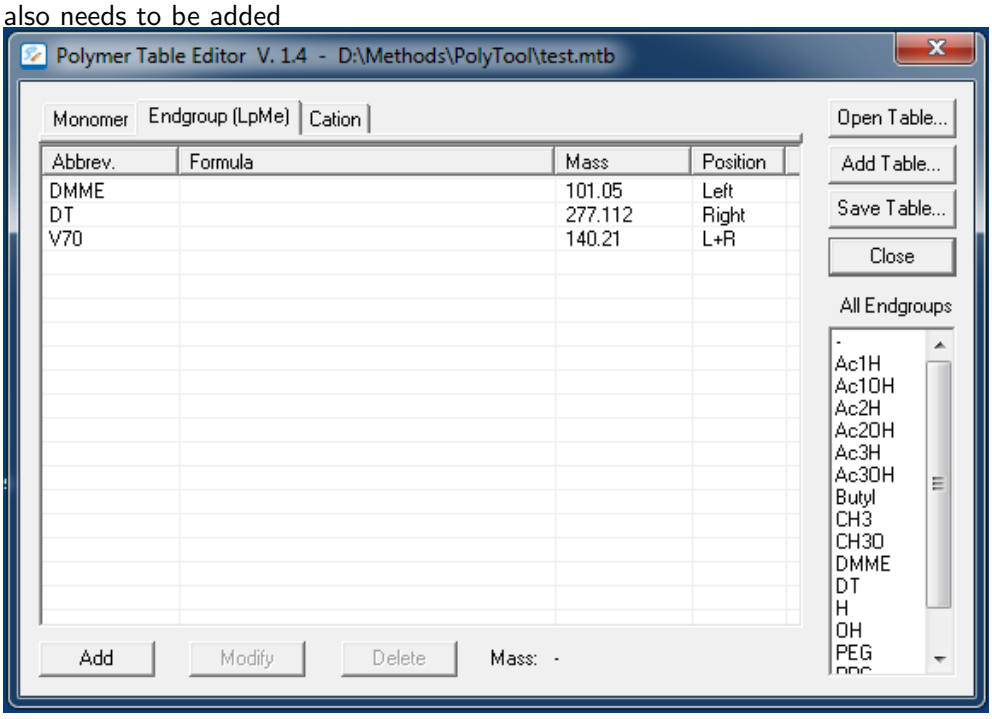
b) Polymer Table Editor: Create tables for your experiments with the masses for your specific monomers and end groups

i. The program has a list of standard monomers, endgroups and cations that can be edited

ii. In the tab *Monomer* new monomers can be added. By entering the chemical formula the mass is automatically calculated. Alternatively the mass can be entered without the chemical formula



iii. For each monomer, end groups can be defined by clicking on the respective row and the tab *Endgroup (abbrev.) – Add*. The position of the end group in relation to the monomer also needs to be added



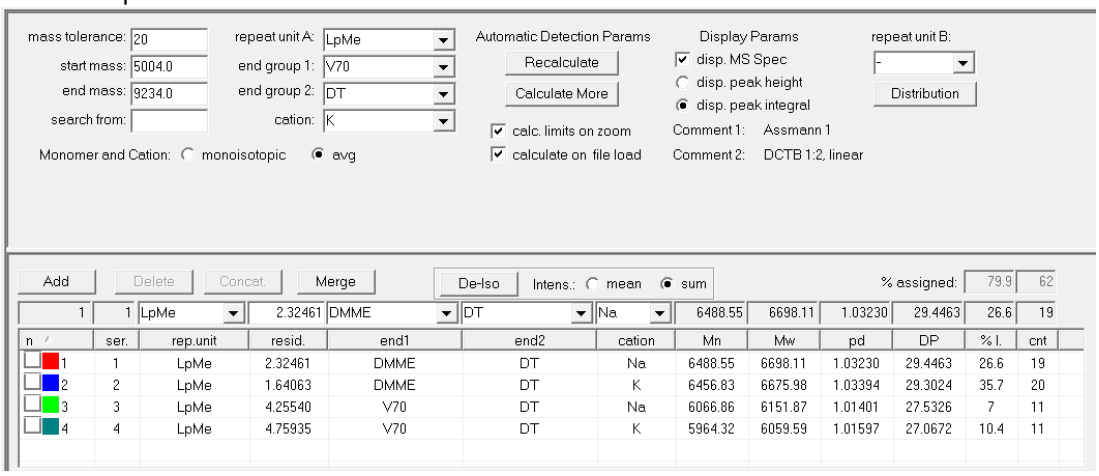
iv. Additional cations can be added or modified in the *Cation* tab if necessary

v. Click *Save Table...* and save it at an accessible location for later use

c) PolyTools: Analyse your polymer mass spectra by assigning the masses to peaks

i. Load the peaklist from *flexAnalysis* and the table from *Polymer Table Editor* into *PolyTools*

ii. The picture below shows the relevant software interface to analyze your polymer MALDI-ToF-MS spectra



A. The table at the bottom shows all the detected signals in which monomers, end groups and cations need to be assigned. The *residuals* are important to assign the correct signals. By checking the boxes in the first column, the entered monomers, end groups and cations become locked and changes cannot be made

B. Monomers, end groups and cations can be varied either at the top or immediately above the table

C. By clicking *Recalculate*, the *residuals* are getting calculated

D. By entering a *repeat unit B*, a second monomer for copolymers can be added

E. If the residuals are similar enough, signals can be *Merge*d

2.1.4 DLS

Dynamic light scattering (DLS) is a method for determining the hydrodynamic radius (r_h) of nano- and microparticles in solution. The measurements can be done on the instrument within the Chair of Physical Chemistry I and according to the instructions that can be provided by the Chair of Marcomolecular Chemistry at HHU. Adding to said instruction, it is important to export the data separated by a tab instead if a comma.

For this work, the raw data and the size distribution have been exported. The raw data entails some of the measurement parameters, such as the dynamic viscosity (η), absolute temperature (T), the number of runs and their duration. Additionally, the r_h of the species within the sample and the polydispersity index (PDI) can be found. The size distribution data is necessary for plotting the size distribution. For this, the data of a single run, which is displayed in one row, is copied into Microsoft Excel. The second half of that row entails the intensity data points. These are cut and pasted below the radii. The edited data can then be paste transposed and plotted in OriginPro 2025 (v10.2).

3 Results & Discussion

4 Summary & Outlook

Acronyms

CeMSA@HHU	Center for Molecular and Structural Analytics @ Heinrich-Heine University
D	diffusion coefficient
$Đ$	dispersity
DLS	dynamic light scattering
DMF	*N*,*N*-dimethylformamide
DOSY	diffusion-ordered spectroscopy
η	dynamic viscosity
HHU	Heinrich-Heine University
MALDI-ToF-MS	matrix assisted laser desorption/ionization time of flight mass spectrometry
M_n	number average molar mass
M_w	mass average molar mass
NMR	nuclear magnetic resonance
PDI	polydispersity index
r_h	hydrodynamic radius
SEC	size-exclusion chromatography
T	absolute temperature
THF	tetrahydrofuran

List of Figures

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Bibliography

[1] DeepL Translator, https://www.deepl.com/en/translator (visited on 26/10/2026).

[2] TopSpin | NMR Data Analysis, https://www.bruker.com/en/products-and-solutions/mr/nmr-software/topspin.html (visited on 26/10/2026).

Appendix

AF4